PROTECTIVE COATING OF STRUCTURAL STEELWORK

MES-ST-SPC-0009

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Appendix A

Paint Systems for Structural Steel Application

1. INTRODUCTION

1.1 Scope

This specification covers the general requirements for surface preparation, application and inspection of protective coatings for structural steelwork.

1.2 Purpose

The purpose of this specification is to set the minimum standard for the technical requirements for shop and site application of paint and protective coatings which includes: the surface preparation, priming, application, testing and quality assurance for protective coatings of structural steelwork, handrails and all other associated metal surfaces, for Ma'aden projects.

1.3 Conflicts and Deviations

Any deviation from the requirements stipulated in this document shall go through a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

ADFT	Average Dry Film Thickness
AISC	American Institute of Steel Construction
ANSI	American National Standards Institute
ASTM	American Society for Testing Materials
DFT	Dry Film Thickness
ISO	International Organization for Standardizations
NACE	National Association of Corrosion Engineers
NDFT	Nominal Dry Film Thickness
SSPC	Society for Protective Coating
SIS	Swedish Standard Institution
CUI	Corrosion Under Insulation

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform work or cause work to be performed.
Manufacturer	Manufacturer Party who produces and warrants performance of products, materials, and/or items provided in accordance with contract documents

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the following applicable codes and standards.

3.2.1 Local Codes and Standards (for Saudi Arabia)

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

Saudi Building Code SBC 306: Structural – Steel Structures.

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.2.2 International Codes and Standards

3.2.2.1. American Institute of Steel Construction (AISC)

AISC 360	Specification for Structural Steel Building

3.2.2.2. American National Standards Institute (ANSI)

ANSI Z535.1	Safety Color Code
ANSI Z535.2	American National Standard for Environmental and Facility Safety Signs

3.2.2.3. American Society for Testing Materials (ASTM)

ASTM A123	Standard Specification for Zinc (hot dip galvanized) Coatings on Iron and Steel Products
ASTM A143/A143M	Standard Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement
ASTM A153/A153M	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM D1535	Standard Practice for Specifying Color by the Munsell System
ASTM D3359	Standard Test Method for Rating Adhesion by Tape Test
ASTM D4752	Standard Practice for Measuring MEK Resistance of Ethyl Silicate (Inorganic) Zinc-Rich Primers by Solvent Rub.
ASTM D2200	Standard Practice for Use of Pictorial Surface Preparation Standards and Guides for Painting Steel Surfaces

ASTM E337	Standard test method for Measuring Humidity with a Psychrometer (the Measurement of Wet- and Dry-Bulb Temperatures)
ASTM B695	Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
ASTM A384/A384M	Standard Practice for safeguarding against Warpage and Distortion During Hot-Dip Galvanizing of Steel Assemblies.
ASTM A385/A385M	Standard Practice for Providing High-Quality Zinc Coatings (Hot-Dip)
ASTM A780/A780M	Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM D4285	Standard Test Method for Indicating Oil or Water in Compressed air
ASTM D5064	Standard Practice for Conducting a Patch Test to Assess Coating Compatibility
ASTM D610	Standard Practice for Evaluating Degree of Rusting on Painted Steel Surfaces
ASTM B117	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D3276	Standard Guide for Painting Inspectors (Metal Substrates)
ASTM D4417	Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel
ASTM D4541	Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers

3.2.2.4. International Organization for Standardizations

ISO 2409	Paints and varnishes – Cross- cut test
ISO 2808	Paints and varnishes – Determination of film thickness
ISO 4624	Paints and varnishes – Pull-off test for adhesion
ISO 8501-1	Preparation of steel substrates before application of paints and related products – Visual assessment of surfaces cleanliness Part 1:

	Rust grades and preparation grades of uncoated steel substrates and of steel substrates after overall removal of previous coatings.
ISO 8503-02	Preparation of steel substrates before application of paints and related products – Surface roughness characteristics of blast – cleaned steel substrates – Part-2: Method for the grading of surface profile of abrasive blast – cleaned steel – Comparator procedure
ISO 8503-04	Preparation of steel substrates before application of paints and related products - Surface roughness characteristics of blast-cleaned steel substrates - Part 4: Method for calibration of ISO surface profile. Comparators and for the determination of surface profile - Stylus instrument procedure.
ISO 8504-2	Preparation of steel substrates before application of paints and related products- Surface preparation methods-Part 2: Abrasive blast-cleaning
ISO 11125 series	Preparation of steel substrates before application of paints and related products - Test methods for metallic blast-cleaning abrasives
ISO 11127 series	Preparation of steel substrates before application of paints and related products - Test methods for nonmetallic blast-cleaning abrasives
ISO 19840	Paints and varnishes - Corrosion Protection of steel structure by protective paint systems - Measurement of, and acceptance criteria for, the thickness of dry films on rough surfaces.
ISO 4628-2	Paints and varnishes - Evaluation of degradation of coatings- Designation of quantity and size of defects, and of intensity of uniform changes in appearance Part 2: Assessment of degree of blistering
ISO 4628-4	Paints and varnishes - Evaluation of degradation of coatings- Designation of quantity and size of defects, and of intensity of uniform changes in appearance-Part 2: Assessment of degree of cracking

ISO 4628-6	Paints and varnishes – Evaluation of Degradation of Coatings Designation of quantity and size of defects, and of intensity of uniform changes in appearance- Part 6: Assessment of degree of chalking by tape method
ISO 19840	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Measurement of, and acceptance criteria for, the thickness of dry films on rough surfaces
ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles - Specifications and test methods.
ISO 12944-4	Paints And Varnishes-Corrosion protection of Steel Structures by protective paint systems-Part 4: Types of surface and surface preparation
ISO 12944-7	Paints And Varnishes-Corrosion protection of Steel Structures by protective paint systems-Part 7: Execution and supervision of Paint work

3.2.2.5. National Association of Corrosion Engineers (NACE)

NACE SP0188	Discontinuity (Holiday) Testing of New Protective Coatings on Conducive Substrates

3.2.2.6. Society for Protective Coating (SSPC)

SSPC-SP COM	Surface Preparation Commentary for Steel and Concrete Substrates
SSPC-SP1	Solvent Cleaning
SSPC-SP2	Hand Tool Cleaning
SSPC-SP3	Power Tool Cleaning
SSPC-SP 5/NACE No.1	White Metal Blast Cleaning
SSPC-SP 6/NACE No.3	Commercial Blast Cleaning

SSPC-SP 7/NACE No.4	Brush-off Blast Cleaning
SSPC-SP 8	Pickling
SSPC-SP 10/NACE No. 2	Near-White Metal Blast Cleaning
SSPC-SP 11	Power Tool Cleaning to Bare Metal
SSPC-SP 14	Industrial Blast Cleaning
SSPC-PA 1	Shop, Field and Maintenance Coating of Metals
SSPC-PA 2	Procedure for Determining Conformance to Dry Coating Thickness Requirements.
SSPC A BOOK	SSPC Painting Manual Volume 1: Good Painting Practice
SSPC SYSTEM BOOK	Systems and Specifications SSPC Painting Manual Volume 2

3.2.2.7. National Association of Corrosion Engineers (NACE)

NACE 37507 Corrosion Prevention by Protective Coatings

German Colour Standard RAL 840 HR Standard Colours (Matt)

3.2.3 Alternative Codes

Alternative codes may be used for Protective Coating of Structural Steelwork design if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0001	Steelwork Fabrication
MD-101-SMEM-EG-ST-SPC-0002	Steelwork Erection

4. GENERAL REQUIREMENTS

4.1 Warranty

The expected service life of the organic liquid-coating systems until the first maintenance coating is 7 years minimum or higher as per project requirements & approved by Employer. A warranty period of 5 years (after acceptance date of completed coating work) shall be required for liquid

coating systems. A lower service life or warranty period is acceptable only if approved in writing by Employer. For repair of coating systems no longer under warranty, the applicable duration of the warranty period shall be two years after completion of the applied repair coating.

4.2 Shop Application

All structural steelwork shall be prepared for painting and have the paint system applied before installation.

In all cases, where surfaces will be inaccessible after shop assembly, they shall be prepared and have the paint system applied before assembly is carried out.

Curing times between successive coats shall be in accordance with those recommended by the manufacturer. The Contractor shall test the degree of cure and the coating shall pass the acceptance criteria provided by the paint manufacturer prior to over-coating or transporting items.

All known field weld areas shall be given the specified abrasive blast surface preparation and primed, but have no topcoat applied for a distance of 50 mm from the weld line. After welding of these areas is completed, they shall be prepared and recoated in the original paint system in accordance with the manufacturer's specifications.

The manufacturer's directions for preparation and application of coatings shall be followed to ensure that the durability of the coating system is not impaired.

The Contractor shall submit for approval the full details of the proposed surface preparation and paint systems prior to the commencement of any surface preparation.

4.3 Site Application

Paint shall be stored only in accordance with the manufacturer's instructions.

All materials used for the specific system being applied shall be products supplied by one manufacturer and details of such product shall be submitted for approval before commencement of work.

The contents of cans shall be thoroughly stirred before being poured into paint pots and shall be thinned only in the specified proportions in accordance with the manufacturer's instructions. Finish coats may be applied by spraying except where any overspray is likely to affect finished surfaces or where spraying constitutes a health hazard to workmen in the other areas. Brush and roller

application will require multiple coats to achieve the specified dry film thickness. Brush and roller application shall be used only after approval from Employer.

The Contractor shall complete the application of any one type of paint or each coat thereof, before beginning the next coat on that section. Curing times between successive coats shall be in accordance with those recommended by the manufacturer. The Contractor shall test the degree of cure and the coating shall pass the acceptance criteria provided by the paint manufacturer prior to over-coating or transporting painted items.

In cases nominated as critical, the application of each coat shall be approved before application of the next coat can proceed, in accordance with hold points nominated in the Inspection and Test Plans (ITPs).

All fittings within any given area are to be painted with the same system as the area unless otherwise specified.

4.4 Qualifications and Materials

All surface preparation, coatings application and inspection, shall be carried out by personnel qualified and experienced in that particular field. Contractors shall submit the names of subcontractors to be employed for the specific work together with the brand names of coating materials for approval prior to commencement of any work.

4.5 Handling and Transport

All steelwork and equipment that has been finish coated shall be handled with care to preserve the coating in the best practical condition. Fabric/sleeve covered slings or similar precautions shall be used to protect paintwork during lifting.

Steelwork shall not be handled until the coating has completely cured and dried hard.

Wherever possible, wooden supports in contact with coated steel during transport and storage shall be covered with a soft material to minimize damage to the coating. Hessian bags /PTFE or elastomer strip shall be used during transportation between coated steelwork and holding down chains to prevent damage to the coating.

4.6 Contractor Responsibility

Contractor shall correct, at his own expense, all defects in coating work and all work not conforming to this specification. Contractor's painting supervisor shall acquaint himself with the salient requirements of this specification, and all other information significant to the work about to be undertaken. He shall be required to provide himself with wet and dry-film thickness gauges, surface temperature gauge, nozzle orifice gauge and nozzle pressure gauge and inspect and monitor the work of surface preparation and painting workers under his direction.

Contractor shall maintain a daily log of ambient temperature, relative humidity, work progress, measured paint film thickness and other information pertinent to the work.

Contractor shall ascertain that all necessary health and safety procedures to protect personnel and surrounding environment during on-site field works.

All relevant safety requirements of applicable safety Standards shall be conformed to, particularly requirements for blasting and power tool cleaning, for scaffolding and access platforms, for compressors and compressed air equipment for ventilation and ventilation equipment in confined areas and for personnel protection, protective equipment and clothing.

Where there is a risk of flammable atmosphere being present in a particular work area, the relevant safety authority may impose restrictions on method of surface preparation. Flame heating requirement shall not be used in hazardous areas.

For internal painting/coating of drinking water facilities (tank, vessels, etc), Contractor shall submit evidence that paint system will not in any way contaminate the water.

On completion of the work, Contractor shall remove all paint containers, clothes, blasting abrasives, equipment and similar items from the work site and dispose of properly.

Contractor is responsible for ensuring that all safety conditions specified in this specification are met.

Tarpaulin covers may be needed for job site blasting cleaning and painting. This shall be supplied, installed and removed by Contractor.

5. METAL SURFACE PREPARATION

5.1 Surface Preparation Procedures

Contractor shall prepare written procedures to be submitted for review and acceptance. The Society of Protective coating SSPC Paint Manual (Volumes 1 and 2) ASTM D2200 shall be used as a reference in addition to requirements given in this specification.

5.2 Pre-Cleaning

Special attention shall be given to all weld areas for removal of weld flux, spatter, slivers and other foreign objects before blasting.

All oil & grease contamination shall be removed by solvent cleaning prior to blasting operations. Degreasing procedure shall correspond to the requirements of SSPC-SP 1.

All sharp edges, fillets, corners and welds shall be rounded (min R = 2 mm) by grinding.

Cutting/heat application often produces hard zones on edges of plates and flanges. Grinding is required where such hard zones occur. i.e. when normal blasting is insufficient to obtain the required roughness.

Any major surface defects, particularly surface laminations, scars and scales, likely to be detrimental to the protective coating system, shall be removed by suitable dressing. Where such defects have been revealed during blast cleaning, and dressing has been necessary, the dressed area shall be re-blasted to the specified standard. All welds are to be inspected and defects repaired prior to final blast cleaning of an area.

5.3 Ambient Condition

Surface preparation and coating shall not be carried out in inadequate light or wet weather and shall be carried out such that the surface being coated is free of moisture, wind-borne or blast cleaning dust. The time interval recommended between surface preparation and coating shall follow the coating manufacturer recommendations.

Coatings shall not be applied if:

- The relative humidity exceeds 85%
- The ambient temperature is less than 5°C or

- The metal temperature is less than 3°C above the dew point.

In any case, humidity, ambient and surface temperature conditions at the time of paint application and curing and drying time before application of the next coat, shall be in accordance with the paint manufacturer's recommendations. These conditions shall be recorded in the Inspection Test Record (ITR) by the Contractor and be available for review.

5.4 Type of Cleaning

Prior to painting, all surfaces shall be free from moisture and thoroughly cleaned of any material that will be conducive to premature failure of the paint coating. All weld spatters shall be removed. The method of surface preparation used shall be compatible with the priming coat of paint and shall be one or a combination of the following:

SSPC Code	Cleaning Type	Description
SSPC-SP 1	Solvent Cleaning	Solvent cleaning is a method for removing all visible oil, grease, soil, drawing and cutting compounds, and other soluble contaminants from steel surfaces.
SSPC-SP 2	Hand Tool Cleaning	Hand tool cleaning is a method of preparing steel surfaces by the use of non-power hand tools. It removes all loose mill scale, loose rust, loose paint, and other loose detrimental foreign matter. It is not intended that adherent mill scale, rust, and paint be removed by this process.
SSPC-SP 3	Power Tool Cleaning	Power tool cleaning is a method of preparing steel surfaces by the use of power assisted hand tools. It removes all loose mill scale, loose rust, loose paint, and other loose detrimental foreign matter. It is not intended that adherent mill scale, rust, and paint be removed by this process.
SSPC-SP 5/NACE No.1	White Metal Blast Cleaning	A commercial blast cleaned surface, when viewed without magnification, shall be free of all visible oil, grease, dust, dirt, mill scale, rust, coating, oxides, corrosion products, and other foreign matter. No staining shall be visible on the entire surface area. The coating specification may require an

SSPC Code	Cleaning Type	Description
		anchor pattern of specified depth for lining applications. This anchor pattern, if specified by the coating manufacturer, requires to be adhered to also.
SSPC-SP 6/NACE No.3	Commercial Blast Cleaning	A commercial blast cleaned surface, when viewed without magnification, shall be free of all visible oil, grease, dust, dirt, mill scale, rust, coating, oxides, corrosion products, and other foreign matter. Staining shall be limited to no more than 33% of each unit area of surface.
SSPC-SP 7/NACE No.4	Brush-off Blast Cleaning	A commercial blast cleaning of uncoated or coated steel surfaces by the use of abrasives. These requirements include the end condition of the surface and materials and procedures necessary to achieve and verify the end condition. A brush-off blast cleaned surface, when viewed without magnification, shall be free of all visible oil, grease, dirt, dust, loose mill scale, loose rust, and loose coating. Tightly adherent mill scale, rust, and coating may remain on the surface. Mill scale, rust, and coating are considered tightly adherent if they cannot be removed by lifting with a dull putty knife after abrasive blast cleaning has been performed.
SSPC-SP 8/	Pickling	A method of preparing steel surfaces by chemical reaction, electrolysis, or both. The surfaces when viewed without magnification shall be free of all visible mill scale and rust.
SSPC-SP 10/NACE No.2	Near-White Blast Cleaning	A near-white metal blast cleaned surface, when viewed without magnification, shall be free of all visible oil, grease, dust, dirt, mill scale, rust, coating, oxides, corrosion products, and other foreign matter. Staining shall be limited to no more than 5% of each unit area of surface.

SSPC Code	Cleaning Type	Description
SSPC-SP 11	Power Tool Cleaning to Bare Metal	Power tool cleaning produces a bare metal surface and to retain or produce a minimum 25 micrometer (1.0 mil) surface profile. This standard differs from SSPC-SP 3, Power Tool Cleaning, in that SSPC-SP 3 requires only the removal of loosely adherent materials, and does not require producing or retaining a surface profile.

Cleaning shall not be carried out in proximity to painting operations or near other surfaces susceptible to dust and particle contamination. Openings on machinery, instrument, etc shall be completely sealed before cleaning is started. Precautions shall be taken to avoid damaging of coating on adjacent areas.

5.5 Abrasive Blast Cleaning

5.5.1 General

Abrasive blast cleaning shall be performed with a clean, sharp grade of abrasive.

Grain size shall be suitable for producing the specified anchor profile. All abrasives shall be free of dust, dirt, and other foreign matter and shall be kept dry at all times.

Silica sand shall not be used as an abrasive blasting material.

If automatic blasting equipment is used, the abrasive mix shall be maintained so that a consistently sharp profile is produced.

Air supply compressor shall be equipped with adequately sized and properly maintained oil and water separators. Compressor shall be capable of maintaining 690 kPa minimum at the nozzle. Additionally, air compressors shall be equipped with either dryers or after coolers to remove entrained moisture from the compressed air.

The presence of oil and water shall be determined in accordance with ASTM D4285 at least two (2) times per twelve (12) hours and following every compressor start-up.

Abrasive blast cleaning shall not be performed when the ambient or the substrate temperature is less than 3°C above the dew point temperature. Dew point temperature shall be determined by

sling psychrometer in accordance with ASTM E337, at the location where work is being performed. Substrate temperature shall be determined with a surface thermometer.

Blast-cleaned surfaces that show evidence of rust bloom or that have been left uncoated overnight shall be re-cleaned to the specified degree of cleanliness prior to coating.

All visible burrs, laminations, slivers, and scabs shall be removed or repaired after blasting.

After blasting and immediately before spraying, dust and loose residues shall be removed by brushing, blowing off with clean dry air, or vacuum cleaning.

Inhibitive washes intended to prevent rusting after blasting shall not be used. Alternative forms of surface preparation shall not be used without written approval.

5.5.2 Blast Cleaning Equipment

5.5.2.1 Pneumatic Systems

The compressed air supply used for abrasive blasting shall be free of water and oil. Adequate separators and traps shall be provided, and these shall be kept emptied. Pressure and capacity should be at least 760 kPa at the nozzle and minimum 10 m3 per minute per nozzle respectively for a maximum nozzle size of 11 mm.

5.5.2.2 Centrifugal Systems

Centrifugal blast cleaning systems are acceptable where they offer economies over pneumatic blast cleaning, provided that:

- The blast profiles and cleanliness standards specified herein are obtained
- Proper precautions in the system design are taken to ensure that shading of recessed or concave areas is prevented. Where it is not possible to provide 100% surface coverage, manual touch-up with pneumatic blasting is an acceptable remedy.
- The abrasive media shall be inspected before, during and after each lot of work piece is completed to verify that no oil contamination of the media has occurred.
- The contaminated abrasive media shall not be reused. In addition, reuse of abrasive media shall also follow coating manufacturer recommendations.

6. COATING APPLICATION

6.1 General

The application method and type of equipment used shall be suitable for the paint specified and the surface being painted. The application method and equipment type shall also follow coating manufacturer recommendations.

Paints and thinners shall be brought to the point of usage in unopened original containers bearing the manufacturer's brand name, color designation and expiry date and ready-mixed unless otherwise specified. Two-pack systems shall be mixed at the site of application to the paint manufacturer's recommendations. The mixed amount prepared shall be no more than the amount that can be applied during the stated pot life.

Color of coating shall be provided by Employer before start of coating work.

Paint shall be applied so that an even film of uniform thickness, tint and consistency covers the entire surface and is free of all defects, including pinholes, runs, sags or excessive brush marks. Film finish shall be equal to that of first-class brushwork.

Where two coats of the same paint are applied, the first coat shall be of lighter tint than the second coat.

Paint ingredients shall be kept properly mixed during paint application.

Equipment shall be kept clean to ensure dirt, dried paint and other foreign materials are not deposited in the paint film. Any cleaning solvents left in the equipment shall be completely removed before painting.

6.2 Coating System Selection

A list of approved coating systems are provided in Appendix A. Contractor may propose for approval alternative coating systems.

Coatings containing cadmium or lead or any material toxic to environment/personnel shall not be used. Contractor shall be completely responsible for determining the compatibility of coating materials being applied to the same item that are the products of different manufacturers.

6.3 Atmospheric Conditions

Painting shall not take place during adverse weather conditions or when these conditions are expected prior to the paint drying. Paint may be applied indoors at any time, provided that the conditions listed below are not present. Generally, paint shall not be applied while any of the following conditions are present:

- **Temperature:** Ambient, substrate or coating material temperatures are above 49°C, below 10°C, or outside the range recommended by the coating manufacturer
- **Humidity:** The ambient or the substrate temperature is less than 3°C above the dew point or outside the range recommended by the coating manufacturer. Dew point temperature shall be determined by sling psychrometer in accordance with ASTM E337, or equivalent electronic device, at the location where work is being performed.
- **Wind:** During strong or gusty conditions, particularly for spray application and when the wind can cause dust or grit on the paint surface
- **Weather:** During rain, or on damp surfaces or surfaces that may have frost. Also, forecast of rain or drizzle may affect the curing period of paint.

6.4 Mixing and Thinning

Materials shall be mixed and thinned in accordance with the coating manufacturer's written instructions.

All mixing shall be done in clean containers that are free from traces of grease, paints, and other contaminants. Containers shall be kept covered to prevent contamination by dust, dirt, or rain.

Mixing of partial kits is not allowed.

Inorganic zinc primers and all pigmented coatings shall be strained prior to application in accordance with the coating manufacturer's written instructions.

Thinning shall not exceed the maximum allowable VOC (volatile organic compound) level for the coating involved.

6.5 Shelf and Pot Life

Materials that have exceeded the coating manufacturer's recommended shelf life shall not be used. No material shall have been in stock for more than 12 months. All paint materials shall be used on a first-in, first-out basis.

The coating manufacturer's recommended pot life shall not be exceeded. When this limit is reached, the spray pot shall be emptied and cleaned, and new material shall be mixed.

6.6 Priming

Primers shall be applied before flash rusting or other contamination occurs.

Previously coated surfaces shall be protected from contamination and overspray.

Primer shall have a uniform thickness over welds, edges, and corners and shall be free of mud cracking.

Primer shall not be applied within 100 mm of unblasted surfaces or 50 mm from the edges to be field welded.

Inorganic zinc primer shall be applied using an agitated pot.

6.7 Top Coating

Each coat of paint shall be of a contrasting color to indicate extent of coverage.

Each coat of multiple coat systems shall be allowed to cure sufficiently in accordance with the coating manufacturer's directions prior to top coating. The minimum drying time between coats shall be in compliance with the manufacturer's instructions. When the maximum recoat, time has been exceeded, previously applied coatings shall be roughened prior to top coating or treated in accordance with the manufacturer's directions.

Prior to top coating over an inorganic zinc primer, the cure of the primer shall be verified. Inorganic primers that are properly cured will polish to bright metal without powdering or loss of material when firmly scraped with the edge of a coin.

Inorganic zinc primers shall have all overspray removed with a stiff bristle brush or wire screen and shall be clean and thoroughly cured prior to top coating.

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When spray applying over inorganic zinc primers, a mist coat shall be used to avoid bubbling. The mist coat may be a thinned coat or may be applied by a quick pass of the spray gun prior to applying the full coat but allowing sufficient time for solvent evaporation.

6.8 Method of Application

6.8.1 General

The method of application shall be governed by the Paint Manufacturer's recommendations for the particular coating being applied.

Each coat shall be applied uniformly over the entire surface, runs, sags and drips shall be avoided. Coating shall be free from pinholes, bubbles and holidays. When these occur, they shall be brushed out immediately or the material shall be removed and the surface repainted. Upon completion of fabrication, any damage to paint system shall be repaired.

6.8.2 Conventional or Airless Spray

Spray equipment shall be equipped with accurate pressure regulators and gauges. Spray equipment shall be in accordance with those recommended by the paint manufacturer.

Air from the spray gun shall be clean and dry with no traces of oil or moisture.

Coatings shall be wet on contacting the painted surface. Areas of dry spray shall be removed and the correct system re-applied.

Spray equipment shall be in compliance with the Paint Manufacturer's recommendation e.g. nozzle tips, spray angle, pressure, etc. Airless sprayers are the preferred equipment. The compressed air supply shall be free of water and oil. Adequate separators and traps shall be provided, and these shall be kept emptied.

All equipment for spray application shall be inspected and approved before any application is begun. A mechanical agitator shall be used to keep the paint under mild agitation during application.

6.8.3 Brush Application

When paints are applied by brushing, brushes shall be of a style and quality that will permit proper application of the material. Brushing shall be done so that a smooth coat, as nearly uniform in thickness as possible is obtained. There shall be no deep or detrimental brush marks. (Brush application gives low film thickness).

Stripe coating shall be mandatory for each coat of a system. Paint shall be worked into all corners and crevices. When applying solvent type coatings, care shall be taken to prevent lifting of previous coats. Brush applied paint shall be of the Coating Manufacturer's brushing formulation.

6.8.4 Roller Application

A uniform method of application shall be adopted when painting large areas. The rolling direction shall minimize paint joint build up. Edges and areas subject to possible roller damage shall be brush-painted prior to rolling. Roller application of the first primer coat is not acceptable.

6.9 Thickness of Coatings

The abbreviation used in the specification NDFT, shall read "Nominal Dry Film Thickness". It shall be observed that excessive film thicknesses may be detrimental due to solvent retention and improper curing. Mud cracking of zinc ethyl silicate could be an additional disadvantage connected with excessive film thicknesses and is not acceptable.

The abbreviation ADFT shall read "Average Dry Film Thickness" and is defined as the actual Average Dry Film Thickness measured for any single 9 m2 area surveyed in accordance with SSPC-PA 2 (Paint Application Specification No. 2).

The maximum thickness in any one application shall not exceed that recommended by paint manufacturer.

6.10 Multiple Coat Applications

Before successive paint coats are applied, intermediate coats shall be inspected for surface contamination. The presence of any grease or oil, shall be removed by a suitable solvent and any salt and dirt adhering to the surface shall be removed by scrubbing with a solution of non-toxic detergent (except those prescribed by the manufacturer as "wet-on-wet"). Removal of contaminants shall only be performed after an intermediate coat has had sufficient time to cure.

The surface shall then be pressure hosed or dusted down by brush to disturb and remove deposits not apparent on visual inspection.

- Coatings shall be applied only under the following conditions:
  - The surface has been cleaned and is dry
  - The manufacturer's stated minimum time for re-coat has elapsed

- The manufacturer's stated maximum time for re-coat has not elapsed. If the maximum time has elapsed, then pre-treatment shall be in accordance with the paint manufacturer's recommendations
- Damaged areas in preceding coats have been made good in accordance with this specification.

Generally, wet-to-wet application is not acceptable.

Visual cracking, mud cracking, flaking or blistering of the coating systems, as defined by ISO 4628-2 and ISO 4628-4, shall not be allowed.

No conspicuous discoloration shall be observed in the topcoats exposed to the atmosphere.

6.11 Hot-Dip Galvanizing

The Galvanizer shall not damage the materials by over pickling or by the use of excessively high temperatures in pickling or galvanizing. Each aspect of the process shall comply with the requirements of ASTM A143.

The Galvanizer shall exercise the required care to safeguard against warpage and distortion during hot- dip galvanizing of steel products in accordance with ASTM A384.

After immersion in the molten zinc, the items shall not be subjected to any process of scraping or wiping which may reduce the uniformity or the specified weight of the zinc coating. Galvanizing practice shall comply with the requirements of ASTM A385.

The zinc coating shall be continuous, smooth and uniform in thickness. The zinc coating shall be free of voids, blisters, flux spots, inclusions, dross, acid or black spots. The coating shall be adherent so it will not be removed by any reasonable process of handling or erection. Lumps, globules, or heavy deposits of zinc, which could interfere with the intended use of the galvanized item, shall not be permitted.

All galvanizing shall be carried out by the hot dipping process and conform to the requirements of ASTM A123.

All welding slag shall be removed by chipping, wire brushing, flame cleaning or abrasive blast cleaning where necessary.

For temporary identification, either water soluble marking paints or detachable metal labels shall be used. For permanent identification, figures shall be heavily punched or embossed by the Contractor.

For galvanized items after pickling, the work shall be inspected and any defects that render the work unsuitable for galvanizing shall be repaired. After such repairs, the work shall again be cleaned by pickling.

The coating mass of zinc shall be as specified on equipment data sheets and the drawings and shall be in accordance with ASTM A123. Galvanized coatings shall be tested. Galvanizing coating thickness shall follow as specified in contract documents and as specified by material engineer on the project.

After galvanizing, all material shall be cooled to air temperature in such a manner that no embrittlement occurs.

Galvanized coatings shall be smooth, uniform, adherent and free from stains, surface imperfections and inclusions.

All fixings including nuts, bolts and washers that are required to be galvanized, shall be hot dipped galvanized in accordance with ASTM A153. All nut threads shall be re-tapped after galvanizing and a lubricant applied in accordance with ASTM A153.

High strength fasteners not allowed to be hot dip galvanized, shall be mechanically galvanized as per ASTM B695.

Cold working of galvanized steelwork shall be avoided.

7. DAMAGE AND INACCESSIBLE SURFACES

7.1 Damaged Paint Surface

Repair of damaged painted surfaces and galvanized holding down bolts after erection shall comply with this Clause and shall also follow coating manufacturer repair recommendations:

The treatment shall be:

- Pre-clean the damaged or unpainted areas
- Decontaminate surfaces

- Disc or hand sand to clean bright metal
- Inorganic zinc primers subject to mechanical damage or weld etc shall be power tool cleaned
- Feather back by sandpapering or whip blasting the original coatings surrounding the damaged area over a 50 mm distance. A rough surface shall be obtained on Epoxy coatings.
- Clean surface to remove all dust
- Conduct surface contaminant test
- Build up a new paint system over the affected area with paints equal to those originally used and having the same dry film thickness for each coat. As an exception, damaged inorganic zinc primers shall be repaired with Epoxy organic zinc rich paint and shall be applied within four hours of blast cleaning.
- The new coatings shall overlap the original coating over the 50 mm prepared distance and shall be color matched to the specified color of the original coating.

7.2 Damaged Galvanized Surfaces

Damaged surfaces shall be repaired per ASTM A780 and specific requirements of this specification. The procedure used to repair damaged mechanically applied zinc coating (galvanizing) shall be submitted to the Employer for approval.

Damaged galvanized areas larger than 3 mm in diameter, caused by oxy-cutting, welding or physical impact shall be treated as follows:

- Prepare the surface by removing any weld slag followed by vigorous power wire brushing of the coating surrounding the damaged area over a 50 mm distance. Areas of spot rust, burnt residue, weld spatter, and other solid adherent foreign matter shall be blast cleaned in accordance with SSPC-SP 14. In the event blast cleaning is not practical, mechanical cleaning in accordance with SSPC-SP-11 using non-woven abrasive discs (3M Company or approved equivalent) may be used. Welded joints of galvanized items shall be cleaned with a non-woven abrasive cup wheel prior to feathering with a non-woven abrasive disc.
- Clean surface to remove all dust
- Apply two coats of organic zinc-rich primer to a minimum dft of 100 microns, and
- Apply a suitable aluminium paint in accordance with the manufacturer's instructions to a dry film thickness of 40 µm. The color shall be matched to the surrounding finish color.

Components that have in excess of 30% of area damaged shall be re-galvanized.

7.3 Inaccessible Surfaces

Surfaces that will be inaccessible after erection of other elements of the structure shall be fully painted prior to the installation of the obstructing item.

8. SURFACES NOT TO BE COATED

The following structural steel items shall not be painted unless otherwise specified in the Contract Documents.

- Machined surfaces
- Contact surfaces of slip-critical type bolted joints in structural steelwork
- Field weld margins.

9. INSPECTION AND TESTING

9.1 General

Any damaged or inferior paint caused by poor workmanship during painting or touchup/repair period shall be repaired at Contractor's expense.

All defective work or work not conforming to the specification shall be repaired at Contractor expense.

Material that is warped or distorted to the extent of impairment for the use intended shall be rejected.

The Employer shall have the right to reject any work, tools, materials, testing equipment, personnel protection, staging and scaffolding at all stages of coating application work where these do not conform to the requirements of the contract or this Specification. Any rejected surface preparation or coating application shall be marked and documented. Any rejected equipment, surface preparation and/or applied coating shall be replaced or corrected to a level in accordance with this specification.

All inspection personnel shall be individually certified by a recognized Body, e.g. NACE, Lloyds, DNV, TUV or Velosi, AND to be agreed by the Employer.

All items coated by the Contractor shall be inspected on arrival and damaged areas shall be repaired by Contractor. In case there is damage of more than 5% of the surface areas or finish coat

appearance is not acceptable, the Employer shall in their sole discretion instruct the Contractor to apply an additional finish coat to improve appearance at no cost to Employer.

9.2 Notification

Contractor shall give seven days notice prior to initially starting any work covered by this specification. While work is in progress, Contactor shall give at least four hours notice of any change of schedule.

9.3 Inspection after Cleaning

Contractor, in conjunction with Employer, shall inspect all surfaces after cleaning and shall identify any defects, improper material or workmanship, or other conditions, which in his opinion will affect the satisfactory performance and permanence of his work.

When such defects have been pointed out, no work shall be started until all such faulty conditions have been corrected.

9.4 Wet and Dry-Film Thickness

In cases of inadequate film thickness, Contractor shall be required to recoat the surface at his own expense and shall do so following the Paint Manufacturer's Procedures pertaining to over coating time intervals which may require surface roughening by abrasive blast.

Coating dry film thickness (DFT) shall be measured by means of a thickness meter based on eddy-current or electromagnetic techniques, in accordance with ISO 2808, Measurements shall be randomly and evenly taken, and the number shall be in accordance with the values given below. The number of measurements shall be increased for areas having a difficult configuration with regard to paint application or limitations in accessibility. Special attention shall be given to areas such as welds, edges, corners and parts where coatings are brush applied. For the acceptance of an inspected area the following criteria shall be fulfilled:

1. The dry film thickness (DFT) of the completed coating system shall be equal to or greater than the specified nominal dry film thickness (NDFT).
2. The dry film thickness (DFT) of any individual coating layer shall be equal to or above 80% of the specified nominal dry film thickness (NDFT) for that coat. Further, the number of readings between 80% of NDFT and NDFT shall be less than 20% of the total number of readings taken.

3. The DFT of the total coating system shall be maximum 1.5 x the NDFT, except for areas such as corners where overlaps are difficult to avoid. For such areas a maximum of 2.5 x the NDFT shall be accepted.

Area: Square meter (Minimum Number of Measurements); Up to 1 (5), 1 to 3 (10), 3 to 10 (15), 10 to 30 (20), 30 to 100 (30), above 100 (additional range, corresponding number of measurements to be added)

9.5 Standards and Equipment

Contractor and/or subcontractor shall provide adequate equipment to perform any work undertaken in this specification including equipment for testing and inspection. Costs associated with delay caused by unsuitable equipment shall be to Contractor's account.

The following standards shall apply:

Table of Standards	
Dry Film Thickness Determination	ISO 2808
Adhesion I: Cross-cut test	ASTM D3359/ISO 2409
Adhesion II: Pull-off test	ISO 4624
Surface Preparation	ISO 8501-1
Surface Roughness	ISO 8503-04
Pinhole detection	NACE SP0188
Surface Contamination	ISO 8502-1, ISO 8502-2 and 8502-3

The following equipment is required to perform the inspection and testing described herein:

1. Blast nozzle orifice gauge.
2. Nozzle pressure gauge.
3. ISO surface profile comparator (ISO 8503).
4. Rugotest No. 3.
5. Magnetic dry film thickness gauge.
6. Electromagnetic d.f.t. gauge.

7. Wet film thickness comb.
8. Hygrometer/psychrometer.
9. Air thermometer.
10. Steel surface thermometer.
11. Dew-point calculator.
12. Tape – ASTM D3359/ISO 2409
13. Non-destructive holiday detector.
14. Sharp thin knife.
15. Microscope with light (30 x).
16. Inspection mirror.
17. Adhesion tester (ISO 4624).

Contractor is required to provide his own inspection equipment. All equipment shall be calibrated in accordance with Manufacturer's procedures. Calibration of equipment's shall be checked each day before use.

9.6 Daily Log

9.6.1 General

A daily log containing the following data shall be maintained:

1. Temperature.
2. Relative humidity.
3. Painting progress.
4. Paint film thickness measurement tests.
5. Any other information pertinent to the work.
6. Results on curing test of zinc ethyl.

The log shall be signed by Contractor.

Numbering system for daily logs must be such that should questions arise, complete history for the area in question can be traced back easily. Daily log shall be part of the final documentation.

9.6.2 Deviation

Major deviations to the specification shall be approved prior to start of the painting. For minor deviation, all agreements made during the progress of the work not in conformity with this specification, or agreement supplementing this specification, shall be in writing.

9.7 Testing

9.7.1 Surface Profile after Blasting Cleaning

- Applicable to all blast cleaning
- Testing equipment – Testex Press-O-Film
- Testing standard – N9 Ba – N10 Ba (30-75 microns).

9.7.2 Surface Cleanliness

- Applicable to all blast cleaned surfaces
- Testing standard – SSPC-SP as required by the primer.

Remarks: The surface shall remain at the blasting standard required by the particular primer, throughout application of the primer.

9.7.3 Dry Film Thickness

- Applicable to all painted surfaces, single coats and complete systems
- Testing equipment – magnetic or electromagnetic d.f.t., gauge/ISO 2808
- Testing standard – as specified for each coating and coating system, reference.

Remarks: The requirements from the equipment Vendors and ISO 2808 about calibration and procedure shall be closely followed. The test shall only be performed on paints sufficiently dry to resist impact penetration from the measuring unit.

9.7.4 Ketone Test on Zinc Ethyl Silicate

- Applicable to zinc ethyl silicate primed surfaces prior to over coating
- Testing equipment - methyl-ethyl-ketone solvent (mek)
- Testing procedure – rub mek or zinc ethyl silicate film firmly with a pad for one minute
- Testing standard – ASTM D 4752. Resistance rating shall be 4 or better.
- Requirement – deterioration of the primer means that further curing is necessary.

Remarks: MEK is a highly inflammable solvent and shall be handled carefully.

9.7.5 Adhesion Tests

- Applicable to – all painted surfaces, single coats and complete systems
- Testing equipment – 1) pull-off test, ISO 4624, pull-off instrument and 2) cross-cut test, ASTM D 3359, method b. Knife and tape
- Testing standard-cross-cut test: grades 4 or 5 in ASTM D 3359 are acceptable.

Remarks: Test shall only be performed on dry and fully cured paint films. The test is destructive, and the test areas shall be marked for touch-up.

9.7.6 Pinhole Detection

- Applicable to special products or areas, for instance tank linings, coatings
- Testing equipment – pinhole detector, NACE SP0188
- Testing standard – the applied voltage shall be according to the coating manufacturer's instruction.

Remarks: No pinholes detected by sparks or sounds are acceptable. Guidelines for performing local repairs versus full repainting shall be submitted by Contractor.

9.7.7 Deviation/Non-Conformance

Non-conformance and/or deviations from the corrosion protection specification requirements shall be identified by Contractor. The deviation report shall also contain proposed corrective actions, with plan for corrective actions to be carried out.

All Corrective actions shall be approved by Employer before execution of respective corrective action.

Non-conformances and/or deviations may be based on, but is not limited to paint film thicknesses, surface roughness, surface cleanliness, paint adhesion, pinholes, visual defects, paint systems and Paint Manufacturer.

9.8 Documentation

Non-conformances deviations from the corrosion protection specifications shall be in the fabrication records. The deviation report shall also contain proposed corrective actions, with dates for corrective actions to be carried out.

Deviations are, i.e. excessive paint film thicknesses, surface roughness, cleanliness, paint adhesion, pinholes, visual defects, paint systems, paint Manufacturer, etc.

As-built drawings shall document where the different paint systems are applied and a reference to the actual drawing shall be included in the Daily Log.

If a question is raised on a special area, it shall be easy to find complete history for the area in question, which means that the numbering system for Daily Logs must be established.

Final inspection-commissioning reports of areas to be enclosed.

Appendix A

Painting system for Structural Steel Application

This information is for reference only and shall be confirmed with the respective coating manufacturer before use.

Painting system for Structural Steel Application MD-101-SMEM-EG-ST-SPC-0009						
S.NO	**Generic System Description**	**Application**	**Surface preparation**	**Profile**	**System Details**	**DFT in µm**
1	Polyamide cured high build Surface Tolerant Epoxy Mastic	Structural Steel Indoor exposure (not exposed to UV, chemicals, non-immersed)	Abrasive blasting to min. Sa 2½ (ISO 8501-1) / SP 10 (SSPC)	Surface profile Medium (G) (ISO 8503-2)	1) SP/ST/HS Epoxy Mastic 2) SP/ST/HS Epoxy Mastic 3) **Total** thickness	1) 150 2) 150 3) 300
2	Polyamide cured high build Surface Tolerant Epoxy Mastic with Polyurethane Top coat	Structural Steel exposed to mild atmosperic exposure	Abrasive blasting to min. Sa 2½ (ISO 8501-1) / SP 10 (SSPC)	Surface profile Medium (G) (ISO 8503-2)	1) SP/ST/HS Epoxy Mastic 2) SP/ST/HS Epoxy Mastic 3) Polyurethane Finish coat - Gloss 4) **Total** thickness	1) 150 2) 150 3) 50 4) 350
3	High build Epoxy Polyamide with MIO	Structural steel exposed to marine and/or high humid condition	Abrasive blasting to min. Sa 2½ (ISO 8501-1) / SP 10 (SSPC)	Surface profile Medium (G) (ISO 8503-2)	1) Inorganic Zinc Silicate (IZS) Primer* 2) HB epoxy Polyamide cured with MIO 3) HB epoxy Polyamide cured with MIO 4) Acrylic Polyurethane - Gloss 5) **Total** thickness	1) 65 2) 125 3) 125 4) 50 5) 365
4	Phenolic Epoxy	Structural steel immersed in seawater	Abrasive blasting to min. Sa 2½ (ISO	Surface profile Medium (G) (ISO 8503-2)	1) SP/HS Epoxy phenolic 2) SP/ HS Epoxy phenolic 3) SP/HS Epoxy phenolic	1) 100 2) 100 3) 100

Painting system for Structural Steel Application MD-101-SMEM-EG-ST-SPC-0009						
S.NO	**Generic System Description**	**Application**	**Surface preparation**	**Profile**	**System Details**	**DFT in µm**
		& insulated, with potential for CUI	8501-1) / SP 10 (SSPC)		4) **Total** thickness	4) 300
5	High solids epoxy, amine cured (or) Coal Tar Epoxy Amine Cured	Structural steel - Buried	Abrasive blasting to min. Sa 2½ (ISO 8501-1) / SP 10 (SSPC)	Surface profile Medium (G) (ISO 8503-2)	1) High solids Epoxy, Amine cured 2) High solids Epoxy, Amine cured 3) **Total** thickness	1) 200 2) 200 3) 400
6	Aluminium Paint	Structural steel - Temporary protection	Abrasive blasting to min. Sa 2 (ISO 8501-1) / SP 6 (SSPC)	Surface profile Fine (G) (ISO 8503-2).	1) Oleoresinous general-purpose Aluminium 2) Oleoresinous general purpose Aluminium 3) **Total** Thickness	1) 25 2) 25 3) 50